

Evaluating Algebraic Expressions (Part II) Homework

① $x^2 + 1 - x$, $x = -3$ 13

② $-y^3 + 2y - 8$, $y = -3$ 13

③ $1 - \frac{2-z}{3}$, $z = 14$ 5

④ $3x - 10 - 4x^3$, $x = -2$ 16

⑤ $5(2y+4)^2 - y$, $y = -1$ 21

⑥ $7 - (9z+1)^2$, $z = 1$ -93

⑦ $x^2 - 4x - x$, $x = -2$ 14

⑧ $\frac{6}{y} - 5y + 2$, $y = 3$ -11

⑨ $-(-3+x^2y)^2 + xy$, $x = 2$, $y = -1$ -51

⑩ $\frac{-z-4}{2} + 19$, $z = -4$ 19

⑪ Find P below if $m = 10$ and $v = 13$.

$P = mv$ P = 130

⑫ Find A below if $W = 8$, $L = 1$, and $H = 3$.

$A = 2LH + 2LW + 2HW$ A = 70

⑬ Find V below if $\pi = 3.14$ and $r = 2$.

You may use a calculator and round to the nearest hundredth.

$V = \frac{4\pi r^3}{3}$ V $\approx$ 33.49

⑭ Find P below if $n = 3$, $R = 6$, $T = 1$, and $V = 6$.

$P = \frac{nRT}{V}$ P = 3

⑮ Find d below if $V = 2$, $V_0 = -1$, and $t = 3$.

$d = \left(\frac{V+V_0}{2}\right)t$ d = $\frac{3}{2}$

⑯ Find y below if $m = 4$, $x = -3$, and $b = 7$.

$y = mx + b$ y = -5

⑰ Find y below if $x = 5$ and $m = -2$.

$y = m(x-6) + 4$ y = 6

⑱ Find y if $a = -2$, $b = 3$, $c = 0$, and $x = 2$.

$y = ax^2 + bx + c$ y = -2

⑲ Find y if $h = -3$, $x = -1$, $k = 1$, and $A = 3$.

$y = A(x-h)^2 + k$ y = 13

Check the following solutions. If a solution works, write "yes." If it does not work, write "no." As with all other problems in your homework, show all work.

⑳ $2x - 3x - 9 = 4x + 1$, $x = -1$ -8 $\neq$ -3, No

㉑ $1 - 2 + y = \frac{y}{3} + 7$, $y = 3$ 2 $\neq$ 8, No

㉒ $-3z - 8z + 5 = 9 - 7z$, $z = -1$ 16 = 16, Yes

$$\textcircled{23} -7x + 2x + 5 - 3x = 4x + 2x, x = 0$$

$$5 \neq 0, \text{No}$$

$$\textcircled{24} 4 + 8y - 12y = 12, y = -2$$

$$12 = 12, \text{Yes}$$

$$\textcircled{25} -\frac{z}{9} + 4 = -6, z = 81$$

$$-5 \neq -6, \text{No}$$

$$\textcircled{26} 7x + 1 = 3 - \frac{2x}{5}, x = 2$$

$$15 \neq \frac{11}{5}, \text{No}$$

$$\textcircled{27} -y + 4 = 5y + y - 3, y = 1$$

$$3 = 3, \text{Yes}$$